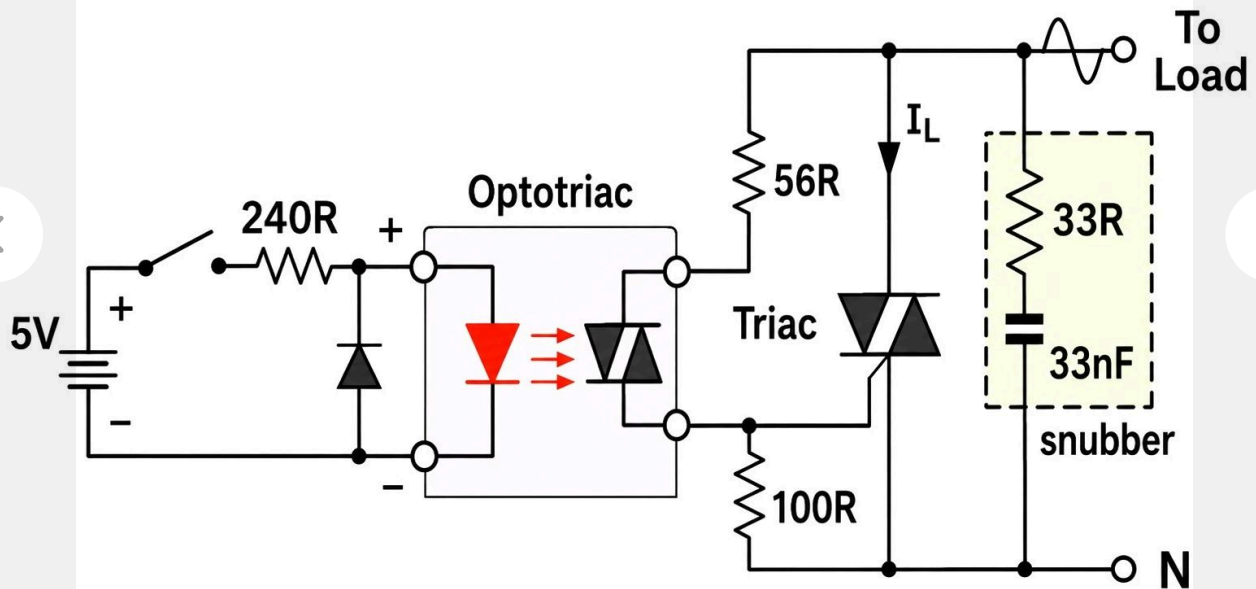




AC LOAD SWITCHING USING OPTOTRIAC



Electronics Education

10 maj kl. 14:17 · 🌐

This circuit shows how an optotriac can safely control an AC load using a low-voltage 5V signal. The left side is the control section, and the right side handles the high-voltage AC power. The key advantage is electrical isolation between the low-voltage electronics and the dangerous AC line.

When the switch is pressed, current flows through the 240Ω resistor and turns on the LED inside the optotriac. That internal light activates the triac driver on the AC side without any direct electrical connection. This isolation helps protect microcontrollers and other sensitive electronics from high-voltage spikes.

The optotriac then triggers the main triac, allowing AC current to flow to the load. Once triggered, the triac continues conducting during each AC half-cycle until the current drops below its holding value.

This makes the circuit useful for switching lamps, heaters, fans, and many AC-powered devices.

The 56Ω and 100Ω resistors help control gate current and improve stable operation. The snubber network made of the 33Ω resistor and 33nF capacitor reduces false triggering caused by fast voltage changes and electrical noise. Snubbers are especially important when switching inductive loads like motors or transformers.

This type of isolated AC switching circuit is widely used in solid-state relays, home automation systems, dimmers, industrial controllers, and embedded electronics projects because it combines safety, reliability, and low-voltage compatibility in a simple design.

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1 v Gilla Svava 2



Jeffrey Bowles

That 33R resistor seems a little low. Maybe more like 100R?

6 d Gilla Svava



Saron Elektronika

you can use SSR too. SSR Solid State Relay

1 v Gilla Svava 1



Dinarajan C N

we can use Quadriac Opto Diac and Triac one chip

1 v Gilla Svava 1



Aiken Biyakna

Pdf?

1 v Gilla Svava



Cristiano Ferreira da Silva

Efter att AI dök upp, började det skapa scheman som alltid var felaktiga, felaktiga värderingar och allt annat. Jag säger inte... Visa mer

1 v Gilla Svava Visa ursprungsversion (Portugisiska) 1

Visa 1 svar

Visa fler kommentarer

6 av 9



Skriv en kommentar ...

